Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Probability & Statistics

BCA — Semester III — 2025 — Answer Sheet

Subject Code: CACS203

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Discuss the application of statistics in computer application.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. The 100 salesmen employed by a company have booked the following number of orders for a newly introduced FAX machine during the last six months:Number of orders Booked10 – 2020 – 3030 – 4040 – 5050 – 6060 – 7070 – 80No. of salesman41225301586Calculate mean, median and mode of the above data.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Define correlation. A school teacher believes that there is a linear relationship between the verbal test score (y) for eighth graders and the number of library books checked out (x). Following are the data collected on 8 students.X121537105229Y7785485975419465(i) Compute the correlation coefficients r between X and Y and Interpret it.(ii) Find coefficient of determination and interpret it.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define regression. The technician now varies the temperature (°C) while keeping other conditions as constant as possible and obtain the following results:Yield (Y)127128130131133Temperature (X)7075808590(i) Construct the regression line of yield on temperature(ii) Predict the yield when temperature is 95.(iii) Interpret the regression coefficients.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define probability. The odds against of A solving a problem as 8 to 6 and the odds in favor of B solving the same problem are 14 to 10. What is the probability that (i) Both A and B will solve it and (ii) A solves it but B fails to solve it?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Write difference between parameter and statistic.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is sampling distribution? Construct frequency distribution table of sample mean in population 2, 4, 6, 8. (i) Write down all possible sample size of two without replacement. (ii) Show that sample mean is an unbiased estimate of population mean.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. What are different methods of measuring dispersion? Following are the marks of Basic Statistics obtained by two students A and B in 9 tests of 100 marks each.Test123456789Mark of A508273452580725662Mark of B456555506862725256(i) Who is better? (ii) If the consistency of performance is the criteria for awarding a prize, who should get the prize?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. The mean weight of products is 68.22 grams with variance of 10.8 grams. How many products in a batch of 1000 would you expect (a) to be over 72 grams, (b) between 70 and 72 grams, and (c) below 65 grams?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. A part of the investigation of the collapse of the roof of a building, a testing laboratory is given all the available bolts that connected the steel structure at 3 different positions on the roof. The forces required to see each of these bolts are as follows:Position 1908279988391Position 2958375102959092Position 383898094Perform an analysis of variance to test at the 0.05 level of significance whether the differences among the sample means at the three positions are significant.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
